Answer 02

Given,

Hypothesis	Error Rate	Errors
H1	6.67%	3
H2	8.89%	4
H3	13.3%	6

Here, we need to determine how confident we are that h2 and h3 perform worse then h1.

Difference in error rates = 8.89 − 6.67 = 2.22%.

Using the standard error of the difference between two proportions, we get −

Standard error = 0.0564

Z-value = $\frac{0.0222}{0.0564} = 0.39$

This gaves approx 65% confidence that h2 performs worse than h1.

Comparasion of H3 with H1

Difference = 13.3 − 6.67 = 6.63%

Standard error = 0.0629

Z-value = $\frac{0.0663}{0.0629} = 1.05.$

This gave approx 85% confidence that h3 performs worse then h1.

we have approx 65% confidence that h2 performs worse than h1 and 85% confidence that h3 perform worse then h1. Therefore, our confidence is higher for h3 because error ~~rate~~ is greater then h1.